

# Product Laws of Exponents

## Lesson Plan

### 1. Lesson Information

|        |                                      |              |              |
|--------|--------------------------------------|--------------|--------------|
| Course | Grade 7 Accelerated<br>(1205050)     | Unit         | 3            |
| Lesson | Lesson 1 — Product Laws of Exponents | Date         | _____        |
| Length | 53 minutes (one period)              | Benchmark(s) | MA.7.NSO.1.1 |

### 2. Standards

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

**Must.** Whole-number exponents with rational bases. Build each law from expansion, then apply it to rewrite an expression and to evaluate one.

**Must not.** No zero exponent yet — that is 3.02 — and no negative exponents until 3.04. No variables until the Thread A days.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR            | What it asks                                                                        | The evidence in this lesson                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts.    | Notes II — $3^2 \cdot 3^3$ is expanded into five factors of 3 before the rule is named, so the law is read off the structure rather than memorized. |
| MA.K12.MTR.3.1 | Complete tasks with mathematical fluency.                                           | Boards 1, 3, 4 and 7 — one law, one move, written and up on the cue; today is where the fluency for the rest of the unit is set.                    |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 5 — when the room splits between $4^5$ and $16^5$ , the sixty-second re-vote before the reveal.                                               |

### 4. Learning Target and Essential Question

**Learning target.** I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

**Essential question.** How can the laws of exponents be used to write an equivalent expression?

**Building on.** MA.6.NSO.3.3 — evaluating positive rational numbers with natural-number exponents (Grade 6). Unit 2 — squares and cubes as repeated multiplication.

**Working toward.** MA.8.NSO.1.3 (integer exponents, Lessons 3.04–3.07) and MA.8.AR.1.1 (the same laws with variable bases, the Thread A days after 3.07).

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 14        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 10        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                                          | DOK   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Why does $5^2 \cdot 5^3$ have five factors of 5 and not six? Show the expansion that settles it.                                                  | DOK 2 |
| $(2 \cdot 5)^3$ equals $2^3 \cdot 5^3$ . Test whether $(2 + 5)^3$ equals $2^3 + 5^3$ , and say what the difference between the two situations is. | DOK 3 |
| A student writes $(3^2)^4 = 3^6$ . Which law did they reach for, and why is it the wrong one here?                                                | DOK 3 |

## 8. Checks for Understanding

each board, and what to do about it

| # | What the board asks                                                                                                                                          | Answer                          | If the board is wrong                                                                                                                                                                                                                                                                                                                                              |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $5^2 \cdot 5^3$                                                                                                                                              | $5^5$                           | $25^5$ — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]; $5^6$ — multiplied the exponents [confused with power of a power]                                                                                                                                                                                                                 |
| 2 | A rectangle is $2^4$ centimeters long and $2^2$ centimeters wide. Its perimeter is 40 centimeters. What is the area of the rectangle, in square centimeters? | $2^6 = 64$ square centimeters   | 40 — the perimeter was copied as the answer [the unneeded figure was used]; 20 — the sides were added, not multiplied; 256 — the exponents were multiplied ( $2^8$ ) [confused with power of a power]; 4096 — the bases were multiplied ( $4^6$ )                                                                                                                  |
| 3 | $(3 \cdot 10)^3$                                                                                                                                             | 27,000                          | 90 — multiplied the base by the exponent [misconception: exponent treated as a factor]; 3,000 — applied the exponent to one factor only                                                                                                                                                                                                                            |
| 4 | $(3^2)^4$                                                                                                                                                    | $3^8$                           | $3^6$ — added the exponents [confused with product of powers]; $9^8$ — squared the base as well                                                                                                                                                                                                                                                                    |
| 5 | $4^3 \cdot 4^2$ Which expression is equivalent?                                                                                                              | $D - 4^5$                       | (A) multiplied the bases as well as adding the exponents [7.NSO.1.1 misconception: operating on the bases]; (B) added the bases and added the exponents [7.NSO.1.1 misconception: operating on the bases]; (C) multiplied the exponents — the power-of-a-power move applied to a product [7.NSO.1.1 misconception, B1G-M]                                          |
| 6 | $(2^4)^3$ Which expression is equivalent?                                                                                                                    | $C - 2^{12}$                    | (A) added the exponents — the product-of-powers move applied to a power of a power [7.NSO.1.1 misconception, B1G-M]; (B) raised the base to the outer exponent as well [7.NSO.1.1 misconception: operating on the bases]; (D) raised the inner exponent to the outer one, $4^3 = 64$ [misconception: the outer exponent applied to the exponent, not to the power] |
| 7 | $(-3)^2 \cdot (-3)^1$                                                                                                                                        | -27                             | 27 — the sign was dropped when the odd power was evaluated [7.NSO.1.1 clarification: rational-number bases include negatives]                                                                                                                                                                                                                                      |
| 8 | $\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$                                                                                                | $\frac{1}{32}$                  | $\frac{1}{64}$ — multiplied the exponents, $(1/2)^6$ [confused with power of a power]; $5/2$ — the exponent 5 written over the 2; $1/10$ — the denominator multiplied by 5 instead of raised to the 5th                                                                                                                                                            |
| 9 | Aiyanna says she can use the power of a product law to evaluate $(9 + (-3))^3$ . Is she right?                                                               | No — $(9 + (-3))^3 = 6^3 = 216$ | 702 — distributed the exponent over the sum [7.NSO.1.1: the laws apply to products and quotients, never to sums]                                                                                                                                                                                                                                                   |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

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**ESE / IEP.** Give the four laws on a half-sheet, one worked example each, in the order of Notes II–IV, and let it stay on the desk through the round. The student's job is to name the law before writing, not to recall it.

**ELL.** Base and exponent are the two nouns the whole unit rests on. Label  $5^3$  once on the board — 5 is the base, 3 is the exponent, 125 is the power — and point at the labels on every board that asks which expression is equivalent.

**Enrichment.** Ask for  $(2^3)^4$  and  $(2^4)^3$  side by side and why they agree, then for a single power of 2 equal to  $4^3 \cdot 8^2$ .

## 10. Closure, Vocabulary, Materials, Practice

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**Closure.** Board 9 is written work and it is the exit evidence: the student has to say that the power of a product law needs a product inside the parentheses, not a sum. Read the boards, not the papers.

**Vocabulary.** **base; exponent; power; product of powers law; power of a product law; power of a power law; identity exponent law**

**Materials.** Whiteboards and markers. Calculators are allowed on everything and are useful for the big values in the bank ( $8^8$ ,  $35^4$ ); the laws themselves are done by hand.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate powers — SZQ; Multiplication with exponents — 5SK; Power rule — 8DA; Understanding exponents — BFA.